

Unit 4.2: Sharpening Frequency Domain Filters

Ideal, Butterworth, Gaussian High Pass Filters & Laplacian Filter

Digital Image Processing Lecture
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Based on: Rafael C. Gonzalez and Richard E. Woods, "Digital Image Processing"

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Lecture Agenda

- Introduction to Sharpening and High Pass Filters
- Ideal High Pass Filter (IHPF)
- Butterworth High Pass Filter (BHPF)
- Gaussian High Pass Filter (GHPF)
- Laplacian Filter
- Comparative Analysis
- Implementation and Applications

What is Image Sharpening?

Definition

Sharpening enhances edges and fine details by amplifying high-frequency components in the frequency domain.

Key Applications:

- Medical image analysis (X-rays, CT scans)
- Satellite and aerial imagery
- Document and handwriting recognition
- Scientific microscopy

Why Use Frequency Domain?

- Direct control over frequency components
- Efficient computation via FFT and Easy filter design and testing

Frequency Domain Perspective

Frequency Components

- **Low frequencies:** Represent smooth, gradual intensity variations
- **High frequencies:** Represent edges, boundaries, and fine details

Sharpening Strategy

High Pass Filter = Attenuate low frequencies + Pass high frequencies

Basic High Pass Filtering:

$$G(u, v) = H(u, v) \cdot F(u, v)$$

where $H(u, v)$ emphasizes high frequencies

Ideal High Pass Filter (IHPF): Definition

Mathematical Formulation

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

where $D(u, v) = \sqrt{u^2 + v^2}$ is the distance from the center

Key Characteristics

- Completely blocks all frequencies within radius D_0
- Passes all frequencies beyond D_0
- Sharp, discontinuous transition at cutoff

Ideal High Pass Filter: The Problem

Gibbs Phenomenon

The sharp cutoff in frequency domain causes **ringing artifacts** in spatial domain.

Visible Artifacts:

- Oscillations around edges (halos)
- Overshoot at boundaries
- Reduced image quality
- Not suitable for most applications

Why does this happen?

- Inverse Fourier transform of discontinuous function $\rightarrow$ oscillatory behavior
- Solution: Use smooth transition filters

Butterworth High Pass Filter (BHPF): Definition

Mathematical Formulation

$$H(u, v) = \frac{1}{1 + \left(\frac{D_0}{D(u, v)} \right)^{2n}}$$

Parameters:

- D_0 = cutoff frequency (radius)
- n = filter order (typical range: 1-5)
- $D(u, v) = \sqrt{u^2 + v^2}$ = distance from center

At Cutoff Frequency:

$$H(D_0, D_0) = \frac{1}{1 + 1} = 0.5 \quad (\text{always, for any order } n)$$

Understanding Filter Order

Effect of Order Parameter n

n	Transition Profile	Ringing Effect
1	Very smooth	Minimal
2	Smooth	Low
3	Moderate	Moderate
5+	Steep (ideal-like)	Higher

Key Trade-off:

- Higher $n \rightarrow$ sharper cutoff (better sharpening)
- Lower $n \rightarrow$ smoother (fewer artifacts)
- **Optimal choice:** $n = 2$ or 3 (balanced performance)

Butterworth HPF: Advantages

Why Choose Butterworth?

1 Smooth Transition

- No discontinuities or sharp jumps
- Gradual frequency response

2 Tunable Characteristics

- Order parameter for flexibility
- Adjustable cutoff frequency

3 Minimal Artifacts

- Much less ringing than IHPF
- Natural-looking results

The most popular choice for practical applications

Gaussian High Pass Filter (GHPF): Definition

Mathematical Formulation

$$H(u, v) = 1 - e^{-D^2(u,v)/(2D_0^2)}$$

where D_0 is related to the standard deviation σ .

Key Points:

- Complement of Gaussian Low Pass Filter
- Exponential decay profile
- At cutoff frequency ($D = D_0$):

$$H(D_0) = 1 - e^{-1/2} \approx 0.393$$

Gaussian HPF: Characteristics

Frequency Response

- **Smoothest transition** among HPF filter
- Exponential increase with distance
- No strict cutoff frequency defined
- Natural, physical behavior

Main Advantages

- Minimal ringing and artifacts
- Very natural-looking results
- Excellent for medical imaging
- Noise-tolerant properties

Preferred choice when quality is paramount

Laplacian Filter: Definition

Laplacian Operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Second-order derivative operator

Spatial Domain Kernel

Standard 3×3 mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Frequency Domain: $H(u, v) = -(u^2 + v^2)$

Laplacian Filter: Properties

Frequency Characteristics

- Response proportional to $D^2(u, v)$
- Zero at origin (removes DC component)
- Quadratic increase with distance
- Strongest at highest frequencies

Key Use Case: Edge Detection

- Emphasizes boundaries and discontinuities
- Produces edge maps (bright on dark)
- Useful for segmentation and feature extraction

Laplacian: Applications

Direct Applications

- 1 **Edge Detection** for boundary detection
- 2 **Medical Imaging** - tumor and structure detection
- 3 **Feature Extraction** for texture analysis

High-Boost Sharpening

Combine with original for enhanced sharpening:

$$g(x, y) = f(x, y) + k \cdot \nabla^2 f(x, y), \quad k > 0$$

Important

Amplifies high-frequency noise. Requires preprocessing filtering.

Filter Comparison: Basic Properties

Property	IHPF	BHPF	GHPF
Smoothness	Poor	Good	Excellent
Ringings	High	Moderate	Minimal
Cutoff	Sharp	Adjustable	Soft

Laplacian Filter:

- No smooth transition (2nd order derivative)
- Excellent for edge detection
- Specialized applications

Filter Comparison: Usage and Performance

Aspect	IHPF	BHPF	GHPF
Practical Use	Limited	Very Common	Very Common
Edge Detection	Fair	Fair	Fair
Sharpening	Good	Excellent	Excellent
Computation	Fast	Moderate	Fast

General Recommendation:

Use BHPF (order 2-3) or GHPF for most applications

When to Use Each Filter

Filter Selection Guide

- **Ideal HPF:** Theoretical analysis, educational purposes
- **Butterworth HPF:** General sharpening, adjustable transition needed
- **Gaussian HPF:** High-quality results, medical imaging, noise-sensitive
- **Laplacian:** Edge detection, feature extraction, boundary analysis

Quality vs. Computational Trade-off

- **IHPF:** Fastest computation, poorest results
- **BHPF:** Moderate computation, good results
- **GHPF:** Moderate computation, excellent results
- **Laplacian:** Fast computation, specialized use

Implementation: Basic Steps

General Procedure:

- 1 **Preprocessing:** Read image, zero-pad
- 2 **FFT:** Apply 2D Fourier Transform
- 3 **Create Filter:** Build $H(u, v)$ in frequency domain
- 4 **Apply:** Multiply $G(u, v) = H(u, v) \cdot F(u, v)$
- 5 **Inverse FFT:** Get result in spatial domain
- 6 **Normalize:** Scale to $[0, 255]$ range

Key Implementation Tips

FFT Considerations

- Use power-of-2 sizes for efficiency
- Pad image to nearest $2^m \times 2^n$
- Use fftshift to center zero frequency

Parameter Selection

- D_0 : Typically 10-50 pixels
- BHPF order: Use $n = 2$ or 3
- High-boost weight: k in range 0.5-2.0

Important

Sharpening amplifies noise. Use preprocessing filtering if needed.

Real-World Applications

Medical and Scientific

- X-ray image enhancement and analysis
- CT scan sharpening for diagnosis
- Microscopy image enhancement
- Tumor and structure detection

Remote Sensing and Imaging

- Satellite and aerial imagery analysis
- Land feature and boundary detection
- Road and infrastructure mapping

Document and Document Processing

- Text document enhancement
- Handwriting and OCR preprocessing

Expected Results

Output Characteristics by Filter

- **IHPF:** Very sharp edges, visible halos
- **BHPF:** Sharp edges, minimal ringing
- **GHPF:** Natural sharpness, smooth quality
- **Laplacian:** Edge maps, good for segmentation

Best Practice:

- Always test on sample images
- Adjust parameters for application
- Validate quality before production use

Key Takeaways

Summary of Filters

- ① **IHPF:** Sharp but causes ringing artifacts
- ② **BHPF:** Most versatile, good balance
- ③ **GHPF:** Best quality, minimal artifacts
- ④ **Laplacian:** Best for edge detection

Trade-off Principle:

Sharp cutoff vs. Minimal artifacts

Questions & Discussion